IB MATH AA SL

Lesson 14

Quadratic Functions & the Discriminant

Functions · 60-minute lesson

Syllabus SL 2.6–2.7 · Property of Lynda Badmus Education

Do Now — recall

ON YOUR WHITEBOARD

1. Factorise $x^2 - 5x + 6$.
2. Solve $x^2 - 9 = 0$.
3. What shape is the graph of $y = x^2$?

Answers (click to reveal)

1. $(x-2)(x-3)$
2. $x = \pm 3$
3. a parabola (U-shaped)

What we're learning today

LEARNING OBJECTIVES

1. **Recognise**
   the forms of a quadratic and what each tells you.
2. **Use**
   the quadratic formula to solve.
3. **Apply**
   the discriminant $\Delta = b^2 - 4ac$.
4. **Determine**
   the number of real roots from Δ.

SUCCESS CRITERIA — I CAN...

✓ read the y-intercept and shape from ax^2+bx+c

✓ solve a quadratic with the formula

✓ calculate the discriminant

✓ decide how many real roots a quadratic has

Solving quadratics

THE FORMULA

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

a, b, c

the coefficients of x^2, x and the constant

± gives

up to two solutions

OTHER FORMS

Factorised

$y = a(x - p)(x - q)$: roots p, q

Vertex

$y = a(x - h)^2 + k$: vertex (h, k)

Standard

$y = ax^2 + bx + c$: c is the y-intercept

Solve with the formula

QUESTION

Solve $x^2 - 4x + 3 = 0$

using the quadratic formula.

SOLUTION

$a = 1$, $b = -4$, $c = 3$. Substitute:

$$x = \frac{4 \pm \sqrt{(-4)^2 - 4(1)(3)}}{2}$$

Simplify the discriminant (16 − 12 = 4):

$$x = \frac{4 \pm 2}{2}$$

Two solutions:

$$x = 3 \quad \vee \quad x = 1$$

Your turn — solve

ANSWER IN YOUR BOOK

1. Solve $x^2 - 7x + 10 = 0$.
2. Solve $x^2 + 2x - 8 = 0$.
3. Solve $x^2 - 6x + 9 = 0$.

Answers (click to reveal)

1. $x = 5$ or 2
2. $x = 2$ or -4
3. $x = 3$ (repeated)

The discriminant

Δ = b² − 4ac

$$= b^2 - 4ac$$

Δ > 0

two distinct real roots

Δ = 0

one repeated real root

Δ < 0

no real roots

WHY IT WORKS

Δ is what's under the root in the formula.

Positive

√ gives two values.

Zero

the ± does nothing → one root.

Negative

can't √ a negative (no real roots).

Use the discriminant

QUESTION

How many real roots does

$x^2 + 2x + 5 = 0$ have?

SOLUTION

a = 1, b = 2, c = 5. Compute Δ:

$$= 2^2 - 4(1)(5) = 4 - 20$$

Δ = −16, which is negative:

$$< 0$$

So there are NO real roots.

EXAM-STYLE QUESTION

Exam-style question

The equation $x^2 + kx + 9 = 0$ has equal roots, $k > 0$.

(a) Write down the condition on the discriminant. [1]

(b) Find the value of k. [3]

MARK SCHEME — reveal after students attempt (tutor only)

(a) $\Delta = 0$ (A1)

(b) $k^2 - 4(1)(9) = 0$ (M1) $\rightarrow k^2 = 36$ (A1) $\rightarrow k = 6$ $(k > 0)$ (A1)

Common mistakes

Sign of b
In the formula it is −b; watch negatives carefully.

Δ meaning
Δ tells the NUMBER of roots, not the roots themselves.

Equal roots
'Equal/repeated roots' means $\Delta = 0$.

y-intercept
In ax^2+bx+c the y-intercept is c, found at $x = 0$.

Mixed practice

QUESTIONS

1. Solve $x^2 - 2x - 15 = 0$.
2. Find Δ for $2x^2 + 3x + 1$.
3. How many roots has $x^2 + x + 4 = 0$?
4. For $x^2 - 6x + k = 0$ with equal roots, find k.

Answers (click to reveal)

1. $x = 5$ or -3
2. $9 - 8 = 1$ (two roots)
3. $\Delta = 1 - 16 < 0 \rightarrow$ none
4. $36 - 4k = 0 \rightarrow k = 9$

Plenary & exit ticket

KEY FORMULAS

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= b^2 - 4ac$$

EXIT TICKET — before you leave

1. Solve $x^2 - x - 6 = 0$.
2. Find Δ for $x^2 + 4x + 4$.
3. How many roots if $\Delta = -5$?

Answers (click to reveal)

1. $x = 3$ or -2
2. 0 (one repeated root)
3. none

Self-assess:

- need more help
- nearly there
- confident

Homework

SET ON DR FROST MATHS (auto-marked)

Task:
Quadratics & the discriminant.

- Aim for 80%+ before the next lesson
- Re-attempt any flagged questions
- Bring one tricky question to discuss

NEXT LESSON

Lesson 15

Rational & reciprocal functions — graphs with asymptotes.

$$y = \frac{ax + b}{cx + d}$$